

Stochastic Heat Equation in the Real World

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Hi, Nick,

Here is a rather complete references. I will keep maintain this file. Please it for private use.

Here are some references where you find SHE naturally arises from physical world:

1. Random polymer model. [[ACQ11](#)], [[MF14](#)].
2. Kardar-Parasi-Zhang equation. [[KPZ86](#)], [[Hai14](#)], [[Qua12](#)], [[GP15](#)].
3. Parabolic Anderson model: [[CM94](#)].
4. Intermittence: [[ZRS90](#)], [[BC95](#)].
5. Evolution of the universe: [[AMS94](#)], [[BCJL94](#)].

These topics are related to each other. This is a very active and fast-developing field.

As for SHE with non Lipschitz coefficients, here are several other references that may interest you: [[DKZ19](#)], [[Mue91](#)], [[FLN19](#)].

Here is one paper that I came across a few years ago, which is related to your invariant measure project. You can take a look and especially you can learn the literature there. The paper is also in the folder “Refs” [[BOS16](#)].

More to come in the future.

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